

BIT MESRA Invigilation Scheduler

Backend Roster Bootstrapping & API Request Routing Lifecycle

1. Server Startup & Bootstrapping Sequence

- **State Verification:** Server checks if solver_state.json exists to retain current running states.
- **Excel Fallback:** If state is missing, parses '*Faculty List (Emp. Code, Phone No. and E-mail ID).xlsx*'.
- **Data Normalization:** Extracts Empl. Code as string keys, standardizes designations (Professor, Associate, Assistant), and builds default history metrics (imbalance = 0.0).
- **State Lock:** Writes initialized dataset to solver_state.json and syncs sample_config.json.

2. REST API Request Routing

- **GET /api/config:** Reads and returns active configurations to frontend dashboard.
- **POST /api/config:** Receives UI configuration payload, writes changes to disk, and updates solver_state.json to maintain persistence.
- **POST /api/solve:** Entry point for the scheduling solver. Parses payloads into dataclasses (AllocationInput) via load_from_dict, triggers the optimizer, and returns structured schedules (AllocationResult).

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Core Invigilation Solver Optimization Pipeline

3. Target Workload & Initial Guess

- **Designation Scaling:** Computes expected duty weight-hours proportionally based on designation ratio multipliers (e.g. Prof=2.0, Assoc=3.0, Asst=4.0).
- **Historical Correction:** Adjusts targets dynamically by factoring in past workloads (+n hours for overload history).
- **Greedy Search:** Orders exam blocks chronologically and allocates slots to qualified faculty members with the lowest accumulated load.

4. Constraint Validation & Local Search Optimization

- **Hard Constraints Enforcement:** Rejects allocations exceeding 1 shift/day, violating availability overrides, or clashing with PG lecture blocks.
- **Hill-Climbing Local Search:** Performs pairwise duty swaps to minimize fairness imbalances.
- **Fairness Verification:** Calculates Gini inequality coefficient & Jain's Fairness Index to compile diagnostics and return the optimal schedule.